Fundamentals of Programming (CS-114)

Python Installation and Working Environments

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Topic 2 - Python Installation and Working Environments

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RECAP

- Computers, because of their efficiency, performing complex computations, and generalization of programs are used in scientific computation.
- Computer programming converts human understandable code to machine code for the computers and machine code to human readable format for user.
- Python program is a collection of instructions in the form of alphanumeric characters while python is the software installed in your computer to convert python program to machine code.
- Since machine code is hard to code, so we need to learn python programming and write instructions for the computers in a python program and hand it over to python for the rest of the job.
- Python is encompassing the most important parts of data, science, engineering, and IT, therefore, is an important programming language.

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OBJECTIVES

After taking this lecture and studying, you should be able to

- Describe the philosophy of proprietary and free and open source software (FOSS), in your own words.
- Describe the reason of various installers of python and which installer to use, in your own words.
- Describe, in your own words, the reason of different working environments and select any one for your future use.

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THE WHY SECTION

- For famous scientific software like MATLAB, Mathematica, and AutoCAD, there is only one installer provided by the company.
- Each of these software have built in packages *e.g.*, MATLAB has many packages like linear algebra package, statistics package, and differential equations solver package.
- The installer provided by the company not only installs the main software but also installs all the packages.
- Although, python also provides the official installer which install python but not the packages.
- There are many installers of python.

THE WHY SECTION

- Then, which installer to use and how to install the packages?
- Just like the installer, the working environment of MATLAB is designed in such a way that the code editor (the file where we write the code), the file explorer (where folders and files are listed), the output window (where we see the results of computations), and many such facilities are available in a single window *click here to see*.
- But, in case of python, different working environments are available to work in python.
- Then, why are there many different working environments to work in python and which one to use?
- We answer these questions shortly.

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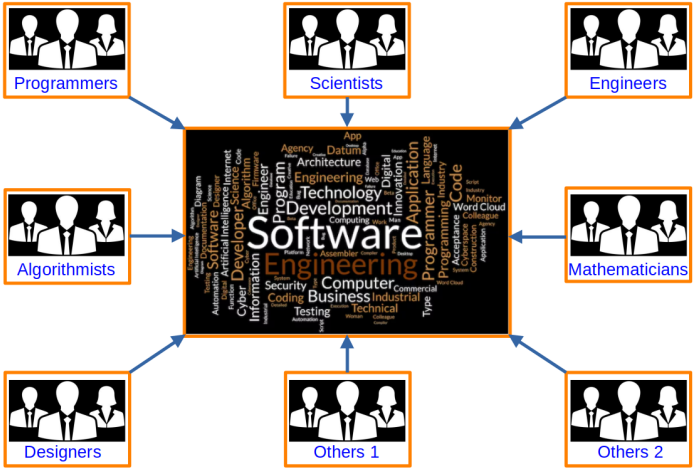
PHILOSOPHY OF FREE AND OPEN SOURCE SOFTWARE

- First we look into the proprietary software.
- A company hires software engineers, relevant scientists, and engineers and develop the software and makes sure to include all the packages and all the working environments in the software.
- The packages, their algorithms, codes and all other relevant information is decided by the personnel in the company.
- The graphic designers work on graphical user interface of the software.
- The final product is released into the market for sale.
- The marketing team works on marketing of the product.

PHILOSOPHY OF FREE AND OPEN SOURCE SOFTWARE


Programmers
Scientists
Engineers
Algorithmists
Mathematicians
Designers
Others 1
Others 2
Software
Engineering
Development
Technology
Engineering
Architecture
Computer
Business
Technical
Testing
Security
Coding
Information
Cyber
Developer
Software
Automation
Program
Engineer
Application
Programmer
Programming
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Science
Data
Business
Digital
Innovation
Cloud
Cybersecurity
Construction
Agency
Application
Industry
Artist
Software
Developer
Engineer
Testing
Automating
Technical
Coding
Security
Computer
Commercial
Business
Industrial
Types
Assembler
Computing
Work
Max
Colleague
Cloud
Word
Monitor
Colleague
Cloud
Word
Agency
Application
Construction
Cybersecurity
Colleague

PHILOSOPHY OF FREE AND OPEN SOURCE SOFTWARE

- There is a main core team which develops the main software and releases it and its versions.
- Different universities and organizations download the source code of the main software and modify according to their needs and requirements and release it and its versions with a different name.
- Sometimes, universities and organizations, instead of downloading the code from the main release, download the source code of some other university or organization, modifies according to its needs and requirements and release it and its versions with a different name.
- Different universities and organizations develop the packages which are usually not available in the main/modified software.

PHILOSOPHY OF FREE AND OPEN SOURCE SOFTWARE

- The packages and its versions are released independently with a different name.
- The packages can be imported in the main/modified releases and used.
- Sometimes, the same computaion can be performed using different packages.
- It is because the code for the computation is released in different packages.
- For example, trigonometric functions may be released in different modified releases as well as packages and we can use any of them.
- A graphical representation on the philosophy of free and open source software (FOSS) is shown in the following figure and the best example of free software is Linux operating system *click here to see*.

PHILOSOPHY OF FREE AND OPEN SOURCE SOFTWARE

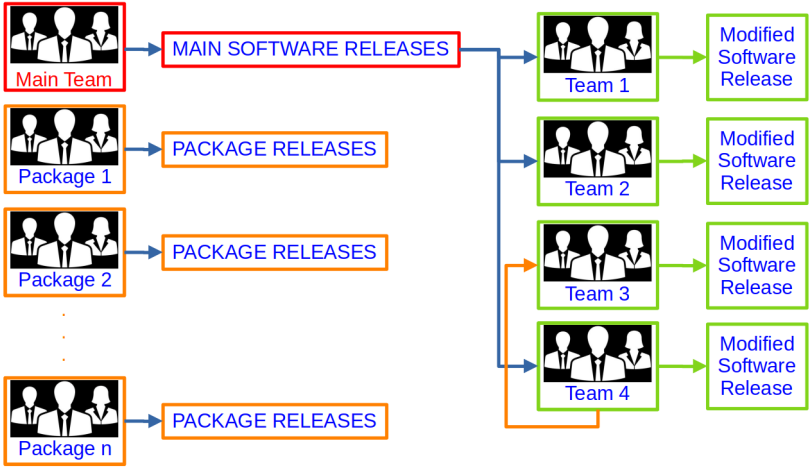

Figure. Philosophy of FOSS.

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PYTHON, PYTHON PACKAGES, AND INSTALLERS

- Python programming languages and its versions are released by Python Software Foundation.
- The installer is released on official webpage of Python Software Foundation *click here to see*.
- There are tens of packages, for various purposes and computations, which can be used with python.
- These packages are not developed by Python Software Foundation and are therefore not released by the company.
- The packages are developed by tens of universities and organizations and knowing about all packages is not easy.

PYTHON, PYTHON PACKAGES, AND INSTALLERS

- Python Software Foundation has developed a reposity with the name of Python Package Index abbreviated as PyPI *click here to see*.
- The packages developed by different universities and organization can be submitted to PyPI from where users can download a package of their interest, install it, import it, and use it.
- If a developed package is not submitted to PyPI and you are aware of it and its developer, you can request the developer of that package to get and use it.
- You can also develop your own packages and submit it to PyPI to help others.

PYTHON, PYTHON PACKAGES, AND INSTALLERS

- Continuum Analytics also famous as Operating System of AI releases python along with several famous packages incorporated in python *click here to see*.
- The company releases python with nearly all packages required in scientific computing *click here to see*.
- It is highly recommended to download the installer for your operating system and install it.
- Once installed, you will hardly need any other package for your computational tasks.
- Downloading python from official website and then installing packages is challenging task if you are new in the world of programming.

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PYTHON WORKING ENVIRONMENTS

- Just like the packages, working environments of python are also multiple.
- There are two modes of working in python namely, 1) interactive mode and 2) non-interactive mode.
- In the interactive mode, python shows output once you complete any line of code if executed.
- In the non-interactive mode, a python program is written and then python is called to translate it to machine code for computer and vice versa.
- The working environments can also be classified into two categories which are 1) text and console arrangement and 2) Integrated Development Environment (IDE).

PYTHON WORKING ENVIRONMENTS

- Text and console arrangement and IDEs fall under the category of non-interactive mode.
- Examples of interactive mode environments are jupyter notebook and google colab.
- IDLE editor with IDLE console and Kate text editor with Konsole are the examples of text and console arrangements.
- The code is written in the text editor and executed in the terminal (console/konsole).
- Pycharm and Spyder belong to IDE class of python where text editor, console, file explorer, and some other features are integrated in the same window.

PYTHON WORKING ENVIRONMENTS

jupyter Untitled Last Checkpoint: 4 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python 3 (ipykernel)

[1]:
```
import numpy as np
import matplotlib.pyplot as plt
```

[2]:
```
x = np.linspace(0, 6.29, 200)
y = np.sin(x)
print(x[:10])
print(y[:10])
```

```
[0.         0.03160804 0.06321608 0.09482412 0.12643216 0.1580402
 0.18964824 0.22125628 0.25286432 0.28447236]
[0.         0.03160278 0.06317398 0.09468208 0.12609559 0.15738313
 0.18851345 0.21945545 0.25017822 0.28065106]
```

[3]:
```
plt.plot(x, y, '-k')
plt.show()
```

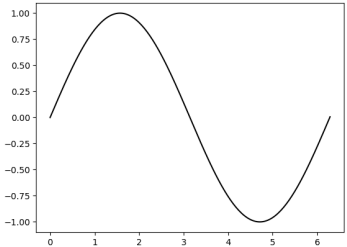


Figure. A snapshot of jupyter notebook

PYTHON WORKING ENVIRONMENTS

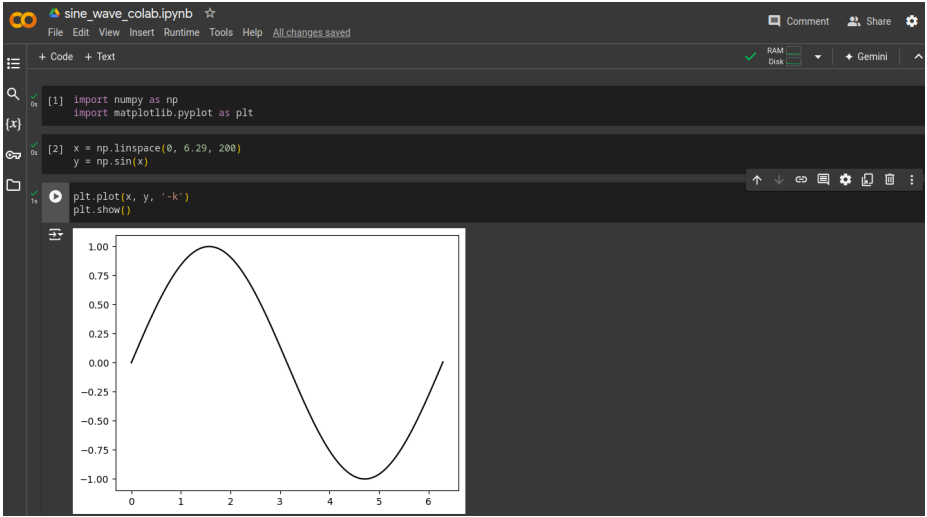


Figure. A snapshot of google colab

PYTHON WORKING ENVIRONMENTS

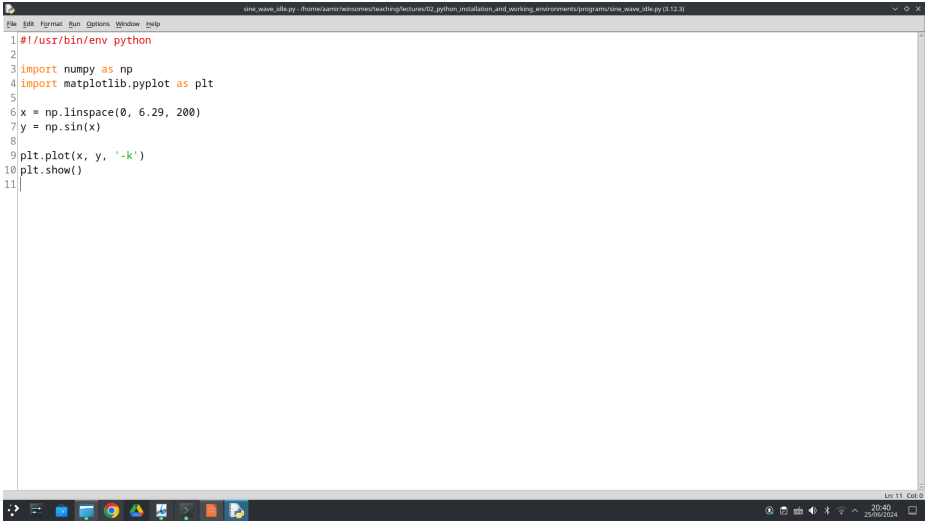


Figure: An example of IDLE code editor

PYTHON WORKING ENVIRONMENTS

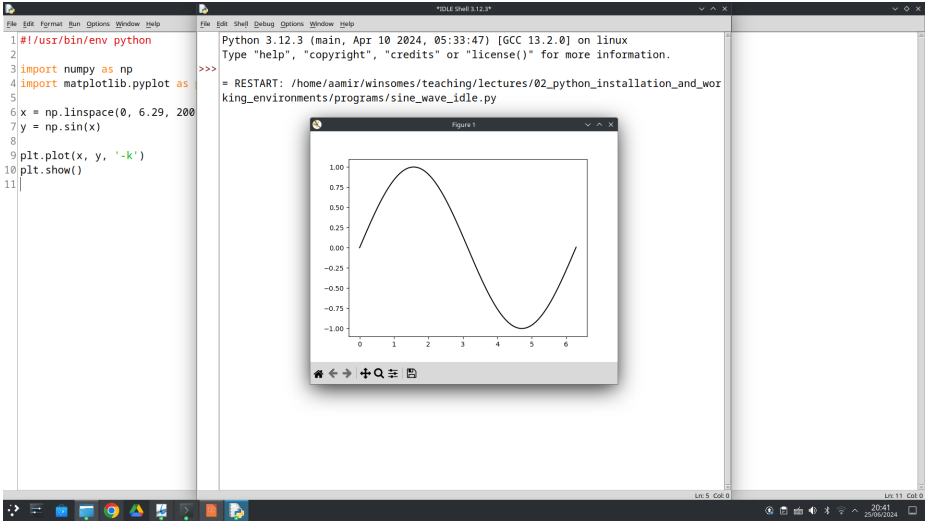


Figure: An example of IDLE environment

PYTHON WORKING ENVIRONMENTS

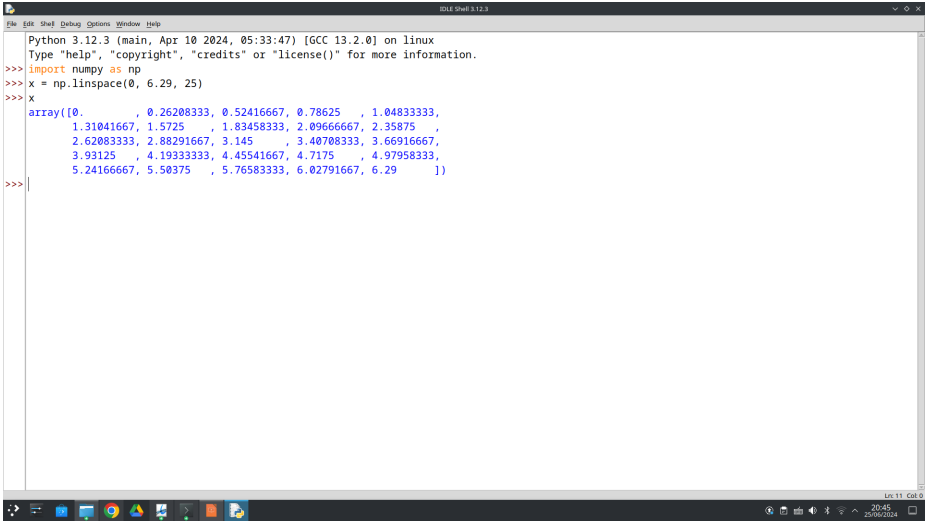
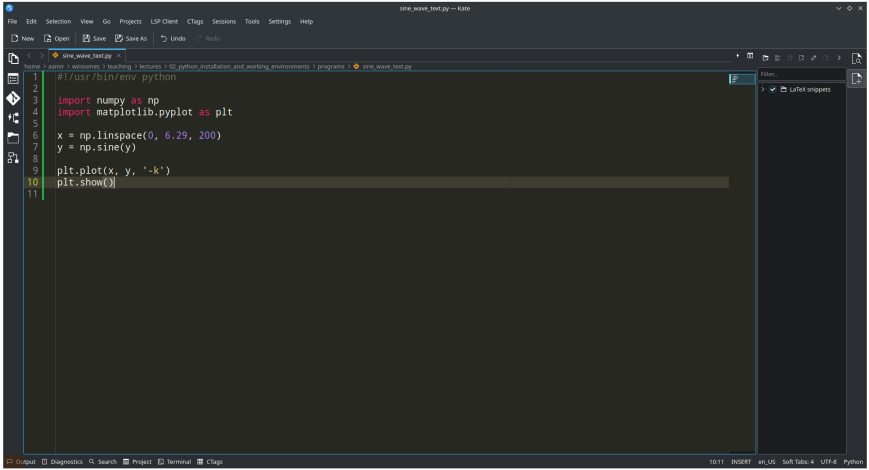


Figure: A screenshot of IDLE [illegible]

PYTHON WORKING ENVIRONMENTS



PYTHON WORKING ENVIRONMENTS

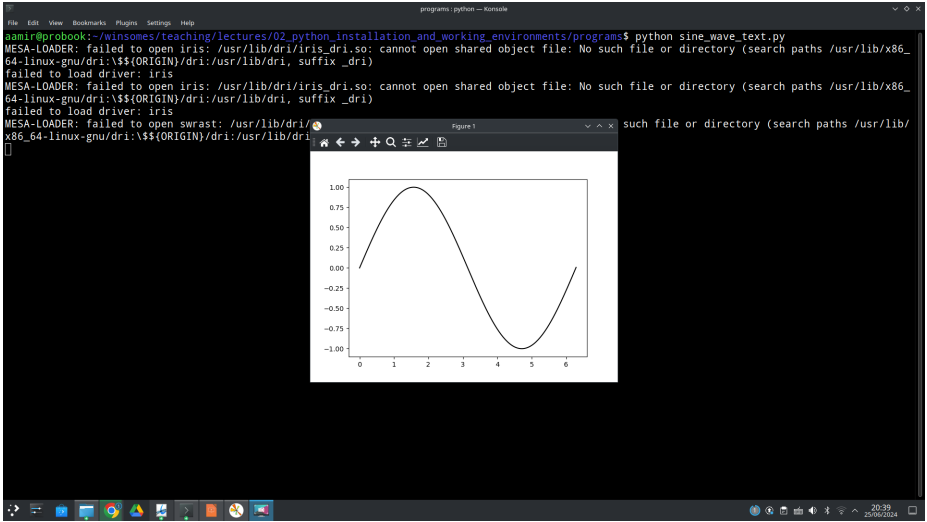


Figure: A screenshot of Kubuntu Konsole (Linux Terminal)

PYTHON WORKING ENVIRONMENTS

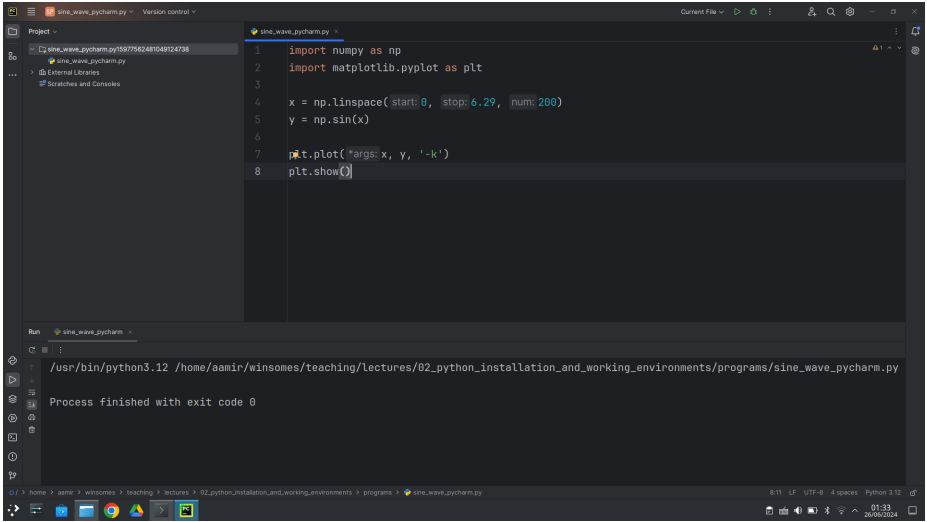


Figure: A screenshot of Pycharm IDE

PYTHON WORKING ENVIRONMENTS

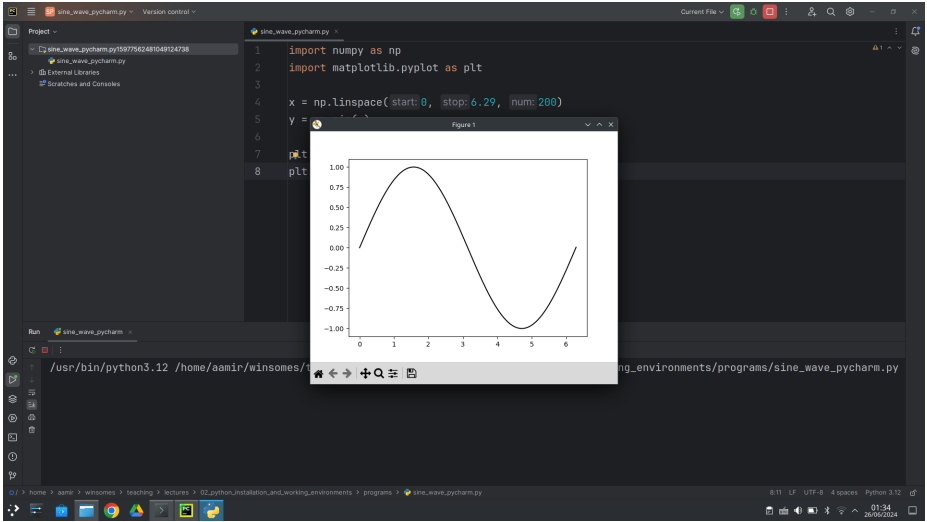


Figure: A screenshot of Python IDE output

PYTHON WORKING ENVIRONMENTS

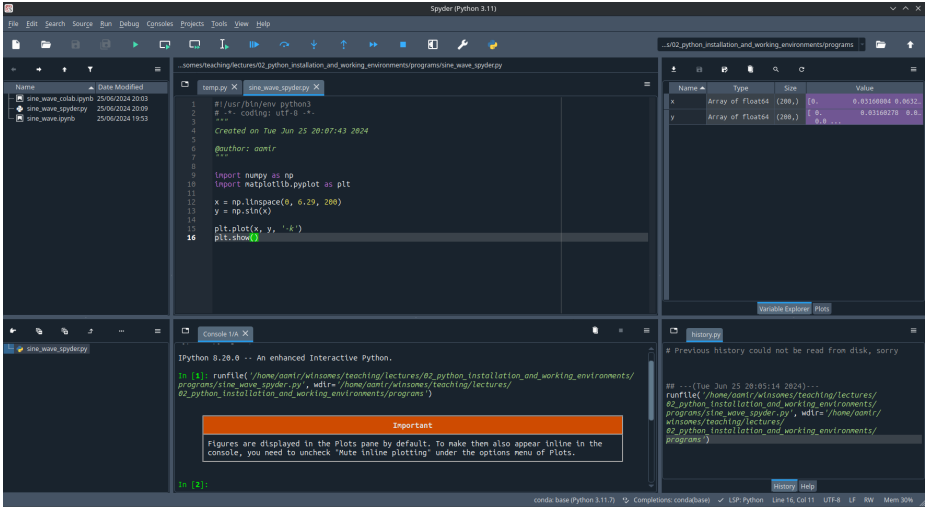


Figure. A snapshot of Spyder IDE (file and variable explorer can be seen).

PYTHON WORKING ENVIRONMENTS

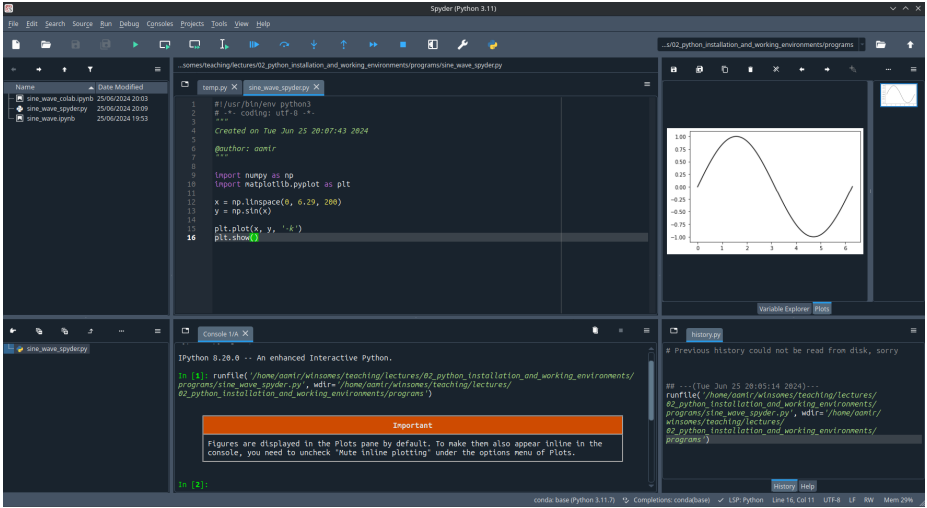


Figure. A snapshot of Spyder IDE (plot explorer can be seen).

PYTHON WORKING ENVIRONMENTS

Figure: A screenshot of a text editor (vim) with IDE

PYTHON WORKING ENVIRONMENTS

Figure: A screenshot of a IDE file editing method

PYTHON WORKING ENVIRONMENTS

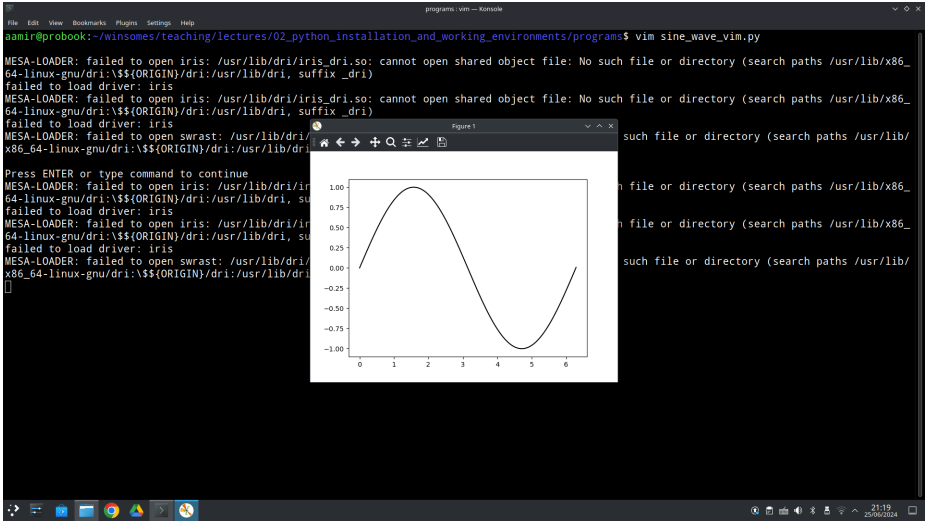


Figure: A screenshot of a IDE output

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SUMMARY

- Python is a free programming language licensed under PSF which can be downloaded, distributed, installed, and used free (libre).
- Python packages are released by PyPI.
- Anaconda provides python with several packages incorporated which is very useful specially for the newbies.
- Python has different working environments including interactive mode, text and terminal, and IDE arrangements.
- The installer provided by Anaconda and Spyder IDE (because it has MATLAB like layout as well) are highly recommended for scientific computing.

THANK YOU!